LITERATURE SURVEY AND PROJECT COMPARISON

EOG-Based Real-Time 3D Target Selection and Ball Control System

Comparison of Three Reviewed EOG Papers with the Current Project

1. Literature Survey – Overview

Electro-oculography (EOG) provides a non-invasive way of converting eye movements into electrical control information. The three reviewed papers address different parts of the EOG problem: (1) understanding eye-movement waveforms and using blink characteristics for drowsiness detection, (2) efficient compression of long-duration EOG recordings, and (3) converting EOG into directional control commands using horizontal and vertical channels. Together, they establish the technical basis for acquiring, conditioning, analyzing and using EOG signals, while also revealing gaps in continuous, real-time, two-axis/3D control.

Literature gap identified

The reviewed work does not provide the complete combination required by the present project: real-time two-axis EOG control, blink-based mode switching/confirmation, continuous ball-position control, 3D target selection using top and side views, and a live graphical interface with a locked final X-Y-Z target.

2. Paper 1 – EOG Study and Drowsiness Detection

Title: EOG based Study of Eye Movements and its Application in Drowsiness Detection

Authors: Pradnya Uday Jambhekar and Sangeetha Prasanna Ram | 2017

Purpose: The paper studies EOG waveforms produced by different eye conditions and applies blink characteristics to detect drowsiness.

- Uses the corneo-retinal potential as the physiological basis of EOG.
- Uses Ag/AgCl electrodes, instrumentation amplification, filtering and ADC acquisition.
- Signal-conditioning chain includes 0.1 Hz high-pass, 30 Hz low-pass and 50 Hz notch filtering.
- The paper reports typical EOG sampling in the 100–400 Hz range and identifies 200 Hz as reliable for its system.
- Different conditions such as left/right movement, normal blink, prolonged blink and consecutive blinks show different waveform characteristics.
- Drowsiness is detected by counting prolonged/drowsy blinks; five consecutive drowsy blinks trigger an alarm.
- The paper emphasizes that measured values vary between individuals, so a fixed threshold should not automatically be treated as universal.

Contribution to the present project: It provides the physiological and signal-conditioning foundation for EOG acquisition and shows why blink events can be used as intentional control events rather than only as a drowsiness indicator.

3. Paper 2 – EOG Compression for Long-Term Recordings

Title: EOG compression in polysomnographic recordings based on the Lempel-Ziv-Welch algorithm

Authors: Alberto López, Gonzalo García, Saeed Mian Qaisar, Francisco Ferrero and Juan Carlos Álvarez | Biomedical Signal Processing and Control, 2025

Purpose: The paper focuses on compressing long-duration EOG data from polysomnographic recordings rather than classifying eye movements for control.

- Uses long-duration sleep recordings containing EOG and other physiological signals.
- Addresses storage and transmission requirements for large EOG/PSG datasets.
- Uses Lempel-Ziv-Welch (LZW), a dictionary-based lossless compression approach.
- The reported dataset contains approximately 8-hour recordings from 22 participants, sampled at 256 Hz.
- The paper evaluates compression ratio/performance, retained energy, PRD, compressed size and runtime.
- Reported LZW results include approximately 84.65% compression performance, 100.44% L2 energy retained and 6.20% PRD.
- A key trade-off is that LZW provides strong compression/reconstruction but is slower than some comparison methods.
- The paper also discusses EOG noise sources such as muscle activity, head movement, blinking and electrode/contact effects.

Contribution to the present project: This paper is less directly related to the control algorithm, but it supports understanding of EOG signal characteristics, noise and the practical importance of efficient digital signal handling.

4. Paper 3 – EOG Eye-Movement and Blink Classification

Title: Detection of Eye-Movement and Blink Patterns using EOG Signals

Journal/Year: Journal of Physics: Conference Series, 2026

Purpose: The paper proposes a low-cost EOG system that classifies four conditions: Blink, Up, Down and Static, using signal processing and a Random Forest classifier.

- Hardware: wet electrodes, Bio Amp EXG Pill, Arduino Uno and computer.
- Uses a 50 Hz notch filter and a low-frequency band-pass; the paper reports 0.5–40 Hz in the methodology and 0.1–40 Hz in the results.
- Normalizes the filtered signal to reduce amplitude differences between recordings/users.
- Divides the signal into overlapping 10-sample windows with a 1-sample step.
- Extracts four features from each window: amplitude, FFT energy, zero crossings and mean.
- Uses Random Forest for classification.
- The original dataset had about 2,000 labelled samples and was expanded to 5,000 through controlled replication/synthetic upsampling.
- The reported accuracy improved from 57.29% to 96.49% after dataset expansion.
- Label smoothing and event segmentation are used to reduce short incorrect prediction changes.
- The paper identifies limitations including offline processing, limited subjects, synthetic data expansion and limited movement classes.

Contribution to the present project: It demonstrates that normalized, windowed EOG features can be used for automated event classification and shows the usefulness of post-processing for turning sample-level predictions into meaningful eye events.

5. Comparative Literature Survey

Aspect	Paper 1 Drowsiness (2017)	Paper 2 LZW Compression (2025)	Paper 3 Classification (2026)	Current Project
Primary objective	Study eye-movement/blink waveforms and detect drowsiness	Compress long EOG/PSG recordings	Classify Blink/Up/Down/Static	Real-time EOG-based 3D target selection and ball-position control
EOG channels	Horizontal + vertical information	EOG in long-term PSG	Mainly vertical movement + blink	Horizontal + vertical channels
Hardware	Ag/AgCl + instrumentation amplifier + PIC ADC	PSG/EOG acquisition system	BioAmp EXG Pill + Arduino Uno + PC	Arduino Uno + 2 BioAmp EXG Pills + ADS1115 + surface electrodes + PC
Signal conditioning	0.1 Hz HPF, 30 Hz LPF, 50 Hz notch	0.9–30 Hz preprocessing described	50 Hz notch + 0.5–40 or 0.1–40 Hz band-pass	50 Hz notch + 0.1–10 Hz band-pass + baseline/drift removal + Kalman smoothing
Main algorithm	Threshold/blink logic	LZW compression	Random Forest	Real-time rule/event detection + state machine + position controller
Window/feature method	Waveform/amplitude/frequency observation	Compression representation	10-sample overlapping windows; amplitude, FFT energy, zero crossings, mean	Continuous streaming; filtering, calibration, baseline tracking and event detection
Control output	Drowsiness alarm	Compressed data	Blink/Up/Down/Static labels	X, Y and Z ball position + target lock
Interaction	Monitoring/alarm	Storage/transmission	Eye-event classification	Interactive GUI for 3D target selection
Real-time focus	Acquisition/analysis system	Long-duration data handling	Mainly offline; real-time proposed as future work	Designed for real-time operation
Major challenge identified	Individual variation and threshold selection	Large data volume and compression/runtime trade-off	Limited data/subjects and classification errors	Drift, cross-talk, blink false detection, continuous control stability

6. Research Gap Identified from the Three Papers

The three papers address complementary aspects of EOG, but none of them combines all of the following requirements into one real-time system:

- Simultaneous horizontal and vertical eye-movement control for a two-dimensional workspace.
- Use of blink events as intentional mode-switching/confirmation commands rather than only as a physiological indicator.
- Continuous ball-position control instead of only discrete four-class labels.
- Separate Top View (X-Y) and Side View (Z) control for three-dimensional target selection.
- Real-time graphical visualization of the ball, target and coordinates.
- A final locking mechanism that confirms the complete X-Y-Z target.
- Online calibration, baseline/drift handling and smoothing suitable for a changing EOG signal.
- A practical control state machine that prevents unintended movement while switching between modes.

Interpretation

The proposed project is therefore positioned as an application-oriented extension of earlier EOG acquisition,

classification and directional-control work. The key distinction is not simply detecting an eye direction; it is using eye events as a real-time control interface for selecting and locking a 3D target.

7. What You Are Currently Making

The current project is a real-time EOG-based 3D target-selection and ball-control system. The user controls a virtual ball using eye movements, while the system separates the task into a Top View for X-Y positioning and a Side View for Z/height positioning.

7.1 Hardware

- Arduino Uno for acquisition/control interface.
- Two BioAmp EXG Pill front ends for EOG signal acquisition.
- ADS1115 16-bit ADC for higher-resolution analog acquisition.
- Surface EOG electrodes for horizontal and vertical eye-movement channels.
- Computer running the real-time Python processing and GUI.

7.2 Signal Acquisition and Processing

EOG electrodes
↓
BioAmp EXG front ends
↓
ADS1115 + Arduino Uno
↓
Serial streaming to Python
↓
50 Hz notch filter
↓
0.1–10 Hz band-pass filter
↓
Dynamic baseline / drift removal
↓
1D Kalman smoothing
↓
Eye-movement + blink detection
↓
State machine
↓
Position controller
↓
3D GUI

Target sampling rate: 250 Hz | **Serial baud rate:** 115200

7.3 Control Concept

- Top View: horizontal and vertical eye movements control the ball position in the X-Y plane.
- Side View: the selected control is used to determine the ball height Z.

- A double blink switches the control from Top View to Side View.
- A triple blink finalizes the selection after the required views are completed.
- After finalization, the complete X-Y-Z target is locked and further movement is disabled.
- The GUI displays the ball, target, coordinates, current mode and status.

8. How the Current Project Extends the Three Papers

Source	What the literature provides	What the current project adds
From Paper 1	Paper 1 establishes EOG acquisition, filtering, waveform analysis and the importance of individual threshold variation.	The current system uses calibration, dynamic baseline/drift handling, filtering and blink events for deliberate control rather than drowsiness detection.
From Paper 2	Paper 2 demonstrates practical handling of long-duration EOG data and highlights noise and data-management considerations.	The current system focuses on low-latency real-time processing rather than compression, so only the signal-handling lessons are directly carried forward.
From Paper 3	Paper 3 demonstrates normalization, windowing, feature extraction, Random Forest classification and event post-processing.	The current system moves toward real-time event/control processing and uses the detected events to control a 3D GUI rather than only producing Blink/Up/Down/Static labels.
Beyond all three	The papers do not combine continuous 3D target positioning, Top/Side mode switching and final target locking.	The current project combines these elements into one interactive EOG human-machine interface.

9. Proposed Contribution / Novelty Statement

Based on the reviewed literature, the project's contribution can be described as follows:

"A real-time EOG-based human-machine interface for three-dimensional target selection, combining horizontal and vertical eye-movement control with blink-based mode switching and confirmation, online signal conditioning, and a graphical X-Y-Z ball-position and target-locking interface."

Important qualification: This statement describes the proposed system relative to the three reviewed papers. A broader literature search would be required before claiming absolute novelty against all published EOG systems.

10. Ready-to-Use Literature Survey Paragraph

Previous studies have demonstrated the usefulness of electro-oculography for eye-movement analysis and human-machine interaction. Jambhekar and Ram (2017) investigated EOG characteristics for different eye conditions and applied prolonged blink detection to drowsiness monitoring, using amplification, filtering and ADC-based acquisition. Their work also highlights that EOG amplitude characteristics vary between individuals, making universal fixed thresholds difficult to apply. A 2025 study by López et al. investigated Lempel-Ziv-Welch-based compression of long-duration EOG recordings in polysomnography, addressing the storage and transmission requirements of large physiological datasets. Kim and Yoon (2013) demonstrated an EOG-based human-machine interface using horizontal and vertical channels to generate four directional commands—left, right, up and down—for a DDR game. Their work also identified DC drift and cross-channel interference as important challenges. More recently, the 2026 study "Detection of Eye-Movement and Blink Patterns using EOG Signals" used normalization, 10-sample overlapping windows, feature extraction and Random Forest classification to classify Blink, Up, Down and Static conditions, reporting 96.49% accuracy after dataset expansion. However, these studies do not combine real-time two-axis EOG control, blink-based mode

switching, continuous 3D target positioning, graphical visualization and final X-Y-Z target locking in one system. The present project therefore focuses on developing a real-time EOG-based 3D target-selection interface using Arduino, ADS1115, BioAmp front ends and Python-based signal processing and GUI control.

11. Current Project – Overall Block Diagram

Eye movements + blinks

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Surface EOG electrodes

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2 × BioAmp EXG front ends

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ADS1115 16-bit ADC

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Arduino Uno

↓

250 Hz serial data

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Python signal processing

↓

Notch + band-pass + baseline/drift + Kalman

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Eye / blink event detection

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Control state machine

↓

X-Y / Z position controller

↓

Top View + Side View GUI

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Final X-Y-Z target lock

12. Final Comparison in One Table

Paper	What it mainly solves	Main limitation relevant to your project	How your project goes further
Paper 1 – Drowsiness	EOG acquisition + eye-condition/blink analysis + drowsiness alarm	Primarily monitoring/drowsiness; fixed-threshold dependence and individual variation	Uses eye events for intentional real-time control and target selection
Paper 2 – LZW Compression	Efficient storage/transmission of long EOG recordings	Not a control/classification system	Focuses on low-latency real-time control instead of compression
Paper 3 – Random Forest	Automatic Blink/Up/Down/Static classification	Offline focus, limited classes and mainly vertical movement	Adds horizontal control, mode switching, 3D positioning and target locking
Current Project	Real-time EOG 3D target-selection and ball-control interface	Main engineering challenge is robust real-time signal/control stability	Integrates acquisition, filtering, calibration, event control, GUI and X-Y-Z locking

13. Conclusion

The three papers provide a progression from basic EOG measurement and physiological event analysis, through EOG data handling, to automated eye-event classification and directional human-machine control. The current project builds on these foundations by targeting a real-time interactive application in which eye movements are converted into controlled X-Y-Z ball positioning and blink events are used for mode selection and final confirmation. The project therefore addresses an application-level gap between simple EOG classification/directional commands and a complete real-time 3D control interface.

14. References Reviewed

1. Jambhekar, P. U., and Ram, S. P., "EOG based Study of Eye Movements and its Application in Drowsiness Detection," International Journal of Advanced Research in Electrical, Electronics and Instrumentation Engineering, Vol. 6, Issue 3, March 2017. DOI: 10.15662/IJAREEIE.2017.0603111.
2. López, A., García, G., Qaisar, S. M., Ferrero, F., and Álvarez, J. C., "EOG compression in polysomnographic recordings based on the Lempel-Ziv-Welch algorithm," Biomedical Signal Processing and Control, Vol. 103, 2025. DOI: 10.1016/j.bspc.2024.107372.
3. Kim, M. R., and Yoon, G., "Control Signal from EOG Analysis and Its Application," International Journal of Electrical and Computer Engineering, Vol. 7, No. 10, 2013.
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